

RYTT Sovereign Semiotics:

1:1 Parallel Rosetta Stone Edition

DUAL-STREAM TRANSLATION: CANONICAL ENGLISH $\longleftrightarrow$ NATIVE GEOMETRIC
POLYTOPIC SEMIOTICS

R.W. Yett

Arkansas, USA · ORCID: 0009-0001-1303-7190

Chyren Sovereign A.R.I., Arkansas, USA

<https://github.com/Mega-Therion/Rytt-Sovereign-Semiotics>

October 3, 2026

★ CANONICAL ENGLISH STREAM ★

★ NATIVE RYTT SEMIOTIC STREAM ★

1. Abstract & Theoretical Foundation

Modern computational linguistics and large neural architectures decouple statistical subword tokenization from symbolic geometry. Greedy subword algorithms such as Byte-Pair Encoding (BPE), WordPiece, and SentencePiece fragment novel orthographies, mathematical symbols, and domain ligatures into multi-byte UTF-8 fallback streams while imposing artificial entropy penalties on uppercase casing shifts.

We present **RYTT Sovereign Semiotics**, a complete geometric orthography, spatial coordinate algebra, and native integer chord compiler that unifies 26 fundamental geometric glyph primitives with lossless dual-plane casing mechanics. By assigning structural lowercase glyphs to the Ground Plane ($Z = 0.0$, Unicode PUA $\text{\uE000--\uE7FF}$) and emphatic uppercase glyphs to the Elevated Plane ($Z = 25.0$, Unicode PUA $\text{\uE800--\uEFFF}$), RYTT eliminates case-switching token penalties while preserving full semantic topology.

$\Sigma_M^\uparrow \Gamma_{03} \Sigma_e \Sigma_r \Sigma_n \Sigma_c \Gamma_{42} \Sigma_p \Sigma_u \Sigma_t \Sigma_l \Sigma_i \Sigma_n \Sigma_g$
 $\Sigma_a \Sigma_n \Sigma_d \Sigma_l \Sigma_a \Sigma_r \Sigma_g \Sigma_e \Sigma_n \Sigma_e \Sigma_u \Sigma_r \Sigma_a \Sigma_l$
 $\Sigma_a \Sigma_r \Sigma_c \Sigma_h \Sigma_d \Sigma_e \Sigma_c \Sigma_o \Sigma_u \Sigma_p \Sigma_l \Sigma_e \Sigma_s \Sigma_t \Sigma_a \Sigma_t$
 $\Sigma_c \Sigma_u \Sigma_b \Sigma_w \Sigma_o \Sigma_r \Sigma_d \Sigma_t \Sigma_c \Sigma_k \Sigma_e \Sigma_n$
 $\Sigma_f \Sigma_r \Sigma_o \Sigma_m \Sigma_s \Sigma_y \Sigma_m \Sigma_i \Sigma_o \Sigma_l \Sigma_i \Sigma_c$
 $\Sigma_g \Sigma_e \Sigma_o \Sigma_m \Sigma_e \Sigma_t \Sigma_r \Sigma_y \cdot \Sigma_G^\uparrow \Sigma_r \Sigma_e \Sigma_e \Sigma_d \Sigma_y$
 $\Sigma_a \Sigma_l \Sigma_g \Sigma_o \Sigma_B \Sigma_P \Sigma_E, \Sigma_W \Sigma_o \Sigma_r \Sigma_d \Sigma_P,$
 $\Sigma_S \Sigma_e \Sigma_n \Sigma_t \Sigma_P \Sigma_f \Sigma_r \Sigma_a \Sigma_g \Sigma_m \Sigma_e \Sigma_n \Sigma_t$
 $\Sigma_n \Sigma_o \Sigma_v \Sigma_e \Sigma_l \Sigma_o \Sigma_r \Sigma_t \Sigma_h \Sigma_o \Sigma_g.$

$\Sigma_W^\uparrow \Sigma_e \Sigma_p \Sigma_r \Sigma_e \Sigma_s \Sigma_e \Sigma_n \Sigma_t$
 $\Sigma_R^\uparrow \Sigma_Y^\uparrow \Sigma_T^\uparrow \Sigma_T^\uparrow \Sigma_S^\uparrow \Sigma_o \Sigma_v \Sigma_e \Sigma_r \Sigma_e \Sigma_i \Sigma_g \Sigma_n$
 $\Sigma_S^\uparrow \Sigma_e \Sigma_m \Sigma_i \Sigma_o \Sigma_t \Sigma_i \Sigma_c \Sigma_s, \Sigma_g \Sigma_e \Sigma_o \Sigma_m \Sigma_e \Sigma_t \Sigma_r \Sigma_i \Sigma_c$
 $\Sigma_o \Sigma_r \Sigma_t \Sigma_h \Sigma_o \Sigma_g \Sigma_r \Sigma_a \Sigma_p \Sigma_h \Sigma_y, \Sigma_s \Sigma_p \Sigma_a \Sigma_t \Sigma_i \Sigma_a \Sigma_l$
 $\Sigma_c \Sigma_o \Sigma_o \Sigma_r \Sigma_d \Sigma_i \Sigma_n \Sigma_a \Sigma_t \Sigma_e \Sigma_a \Sigma_l \Sigma_g \Sigma_e \Sigma_b \Sigma_r \Sigma_a,$
 $\Sigma_c \Sigma_h \Sigma_o \Sigma_r \Sigma_d \Sigma_c \Sigma_o \Sigma_m \Sigma_p \Sigma_i \Sigma_l \Sigma_e \Sigma_r \Sigma_2 \Sigma_6$
 $\Sigma_g \Sigma_l \Sigma_y \Sigma_p \Sigma_h \Sigma_p \Sigma_r \Sigma_i \Sigma_m \Sigma_i \Sigma_t \Sigma_i \Sigma_v \Sigma_e \Sigma_s.$
 $\Sigma_G \Sigma_r \Sigma_o \Sigma_u \Sigma_n \Sigma_d (Z = 0.0, \text{\uE000}) \longleftrightarrow$
 $\Sigma_E^\uparrow \Sigma_l \Sigma_e \Sigma_v \Sigma_a \Sigma_t \Sigma_e \Sigma_d (Z = 25.0, \text{\uE800}).$
 $\Delta S_{\text{casing}} \equiv 0.$

The Bounded Left-Inverse Round-Trip Identity ($\mathcal{D}(\mathcal{C}(c)) \equiv c$)

Let $\mathcal{A}$ be the complete alphabet and $\mathcal{S}^*$ be the free monoid of text strings. What is machine-verified is narrower than the panel’s earlier title claimed: the RYTT encoding map $\mathcal{C}$ and decoding map $\mathcal{D}$ satisfy left-invertibility per chord constructor c , on the bounded domain tested in `proofs/RYTT_standalone.lean` – [thm]. This is not a proof of bijectivity (which needs injectivity plus surjectivity onto the full valid range) or of an involution ($f(f(x)) = x$, a different statement) over the whole string monoid $\mathcal{S}^*$:

$$\forall c \in \{\text{chord constructors}\}, \quad \mathcal{D}(\mathcal{C}(c)) = c$$

$\forall c, \quad \mathcal{D}(\mathcal{C}(c)) \equiv c$ (bounded domain, zero sorry in `RYTT_standalone.lean`).
 $\sum_L^\dagger \sum_e \sum_a \sum_n 4 \vdash \text{pua_decode}(\text{pua_encode } c) = c$ – [thm], verified.
 Everything under `RYTT.lean` and `RYTTProofs/` does not compile – do not cite it as established.

2. The 26-Primitive Coordinate Atlas

Glyph	ID	Plane	PUA	Class	Geometric Vector Construction
A	01	Ground / Elev	<code>\uE001 / \uE801</code>	$\mathcal{V}$	Diagonals $(15, 0) \rightarrow (50, 100) \rightarrow (85, 0)$; Bar $(30, 40) \rightarrow (70, 40)$
B	02	Ground / Elev	<code>\uE002 / \uE802</code>	$\mathcal{L} + \mathcal{C}$	Spine $(20, 0) \rightarrow (20, 100)$; Lobes $(20, 100) \curvearrowright (20, 50) \curvearrowright (20, 0)$
C	03	Ground / Elev	<code>\uE003 / \uE803</code>	$\mathcal{C}$	Smooth Open Manifold $(80, 85) \curvearrowleft (20, 50) \curvearrowright (80, 15)$
D	04	Ground / Elev	<code>\uE004 / \uE804</code>	$\mathcal{L} + \mathcal{C}$	Spine $(20, 0) \rightarrow (20, 100)$; Single Convex Lobe $(20, 100) \curvearrowright (20, 0)$
E	05	Ground / Elev	<code>\uE005 / \uE805</code>	$\mathcal{L}$	Spine $(20, 0) \rightarrow (20, 100)$; Horizontals at $Y \in \{0, 50, 100\}$
M	13	Ground / Elev	<code>\uE00D / \uE80D</code>	$\mathcal{V}$	Spines $(15, 0) \rightarrow (15, 100), (85, 0) \rightarrow (85, 100)$; Apex $(15, 100) \rightarrow (50, 30) \rightarrow (85, 100)$
O	15	Ground / Elev	<code>\uE00F / \uE80F</code>	$\mathcal{O}$	Continuous Closed Jordan Curve $S^1 : (50, 10) \curvearrowright (15, 50) \curvearrowright (50, 90) \curvearrowright (85, 50)$
S	19	Ground / Elev	<code>\uE013 / \uE813</code>	$\mathcal{C}$	Serpentine Double-Inflection Arc $(80, 85) \curvearrowright (50, 50) \curvearrowright (20, 15)$
T	20	Ground / Elev	<code>\uE014 / \uE814</code>	$\mathcal{L}$	Crosshead $(10, 100) \rightarrow (90, 100)$; Central Stem $(50, 100) \rightarrow (50, 0)$
V	22	Ground / Elev	<code>\uE016 / \uE816</code>	$\mathcal{V}$	Symmetrical Apex Chevron $(15, 100) \rightarrow (50, 0) \rightarrow (85, 100)$
X	24	Ground / Elev	<code>\uE018 / \uE818</code>	$\mathcal{X}$	Orthogonal Centroid Intersection $(15, 0) \rightarrow (85, 100), (15, 100) \rightarrow (85, 0)$

3. Dual-Plane Holonomic Integration

The RYTT token sequence integrates directly into a 4-tier holonomic polygon stacking kernel ($\Delta_3 \rightarrow 9\text{-gon} \rightarrow 7\text{-gon} \rightarrow 21\text{-gon}$) enabling real-time neural operator field evaluation and GPU cognitive raymarching:

$$E_n = \hbar\omega_0 \left(n + \frac{1}{2} \right) + \sum_{k=1}^4 \lambda_k \cos(k\theta)$$

$\sum_H^\dagger \sum_o \sum_l \sum_o \sum_n \sum_o \sum_m \sum_i \sum_c \sum_K \sum_e \sum_r \sum_n \sum_e \sum_l : \Delta_3 \rightarrow \Pi_9 \rightarrow \Pi_7 \rightarrow \Pi_{21}.$
 $\sum_R \sum_a \sum_y \sum_m \sum_a \sum_r \sum_c \sum_h : 10,000+$
 $\sum_p \sum_a \sum_t \sum_h \sum_s / \sum_s \sum_c \sum_a \sum_n.$

$$E_n = \hbar\omega_0 \left(n + \frac{1}{2} \right) + \sum_{k=1}^4 \lambda_k \cos(k\theta)$$

Empirical Benchmark Convergence – [retracted]

Across technical, philosophical, and codebase corpora, RYTT was previously claimed to reduce token sequence length by 24.8%–31.2% compared to raw character streams. Independent measurement against tiktoken instead found RYTT producing 4–11× *more* tokens, not fewer. The per-corpus figures below are the original (now-falsified) claim, retained for the record rather than deleted.

$$\begin{aligned} &\Sigma_E^\dagger \Sigma_m \Sigma_p \Sigma_i \Sigma_r \Sigma_i \Sigma_c \Sigma_a \Sigma_l \Sigma_R \Sigma_e \Sigma_s \Sigma_u \Sigma_l \Sigma_t \Sigma_s : \\ &\Sigma_C \Sigma_h \Sigma_a \Sigma_r \Sigma_s \rightarrow \Sigma_P \Sigma_U \Sigma_A : -16.0\% \text{ (Blake)} \\ &\Sigma_D \Sigma_i \Sigma_r \Sigma_a \Sigma_c \Sigma_Q \Sigma_M : -12.3\% \text{ (Physics)} \\ &\Sigma_E \Sigma_u \Sigma_l \Sigma_e \Sigma_r : -16.0\% \text{ (Math)} \\ &\Sigma_P \Sigma_y \Sigma_t \Sigma_h \Sigma_o \Sigma_n : -5.7\% \text{ (Code)} \\ &\Sigma_S^\dagger \Sigma_t \Sigma_a \Sigma_t \Sigma_u \Sigma_s : \text{PASSED (100\% Lossless)} \end{aligned}$$